

AI TDS COMPLIANCE ASSISTANT

Project Summary

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ICAI AI Level 2 Batch	B 84 - Gurugram
Project Type	AI-enabled Tax Deduction at Source (TDS) Compliance Application
Application	TDS Assistant Pro
Live URL	https://tdsassistant.lovable.app
Assessment / Working Year	AY 2026-27 / FY 2026-27 project environment

Purpose of the Project

The project is designed to assist a Chartered Accountant or finance team in converting vendor invoices into a structured TDS register, identifying the likely TDS category, maintaining vendor-wise cumulative amounts, detecting duplicate invoices, and presenting the TDS position in a simple reviewable dashboard.

The emphasis is on reducing repetitive manual work while keeping the user in control of tax-sensitive decisions.

1. Executive Summary

The AI TDS Compliance Assistant is a web-based application created as an ICAI AI Level 2 capstone project. It addresses a common practical problem in professional accounting work: invoice-wise TDS review is often spread across PDFs, spreadsheets and manual calculations. This creates delays, duplicate processing, inconsistent vendor classification and difficulty in monitoring cumulative thresholds.

The application centralises the workflow. A user uploads one or multiple invoices, the system extracts invoice data, stores each document in a TDS register, identifies the payment category from rules and prior vendor history, computes the relevant base amount, tracks cumulative amounts and flags records that need review. Normal text-based PDFs are processed primarily through deterministic parsing, while AI is intended as a fallback for difficult or unstructured cases.

2. Problem Statement

- Manual invoice review requires repeated entry of vendor name, invoice number, invoice date, taxable value and nature of payment.
- A single vendor may have several invoices during the year, making threshold tracking difficult if invoices are reviewed individually.
- Duplicate invoices can result in incorrect cumulative values and incorrect TDS computation if not identified.
- Invoice layouts differ widely; bank account numbers, GST rates, serial numbers or quantity fields can be mistaken for monetary amounts by a weak parser.
- A CA needs an audit-friendly register where the extracted data, cumulative position, TDS result and review status remain visible and editable.

3. Project Objectives

- Create a single TDS working register from uploaded invoices.
- Extract essential invoice fields automatically and quickly.
- Derive the TDS base from invoice value and separately identified GST wherever appropriate.
- Apply configurable TDS rules instead of permanently hard-coding tax decisions inside the invoice parser.
- Maintain vendor-wise cumulative position across the financial year.
- Retain duplicate invoices for audit visibility while giving them zero impact on cumulative/TDS calculations.
- Provide review controls, search, filters and Excel export for practical office use.
- Reduce dependence on paid AI calls by using rule-based parsing first and AI only when required.

4. Proposed Solution and Workflow

Step	System Activity	Result
1	Upload single invoice or multiple PDFs	Files enter processing queue
2	Extract selectable PDF text	Fast processing without AI for normal PDFs
3	Parse invoice fields	Vendor, invoice no., date, description, GST and amounts
4	Validate amounts	Reject percentages, account numbers, HSN codes and unrelated numeric values
5	Derive TDS base	Taxable/base amount or gross less separately identified GST
6	Match vendor / payment type	Use vendor history and TDS Rules master
7	Store in TDS Register	Document becomes visible even if some fields need review
8	Check duplicate status	Duplicate remains visible but has zero financial effect
9	Recalculate cumulative position	Previous and new cumulative are updated vendor-wise
10	Display TDS status	Below threshold / TDS required / duplicate / review status

5. Major Functional Modules

Dashboard

High-level visibility of invoice/TDS activity and working status.

Upload Documents

Upload individual invoices or multiple invoices for batch processing.

TDS Register

Central ledger showing date, vendor, invoice, category, section, base, previous cumulative, new cumulative, rate, catch-up, TDS now, TDS status and review status.

Summaries

Consolidated views useful for review and management reporting.

TDS Rules

Configurable payment categories, sections, rates and threshold logic used by the processing engine.

Search and Filters

Search by vendor, invoice, category, section, TDS status or review status; filter documents and duplicates.

Export to Excel

Export the register for further working, documentation or submission.

Vendor Memory

Reuse previously confirmed vendor/payment classification so the same vendor does not need repeated AI classification.

6. Invoice Parsing and Data Validation

A major development focus of the project was making invoice extraction reliable without using AI for every document. The parser was refined to distinguish genuine monetary values from unrelated numbers commonly present on invoices.

Validation Rule	Purpose
Ignore percentages such as 9%, 18%, 10%	Prevents GST/TDS rates from becoming invoice amounts
Ignore bank account numbers and phone numbers	Prevents very large false TDS bases
Ignore HSN/SAC codes and invoice/serial references	Prevents identifiers from being treated as values
Recognise labels such as Taxable Value, Professional Fees, Hire Charges, Subtotal and Total	Improves amount selection using invoice context
Use arithmetic validation: Base + CGST + SGST + IGST approximates Gross	Cross-checks extracted figures
If GST is separately identified, derive base from gross less GST where necessary	Supports invoices where taxable value is not explicitly labelled
If a critical field cannot be confidently derived	Keep invoice in register and mark Needs Review rather than losing the document

Illustrative Example

Particulars	Amount
Professional / Service Base	Rs. 25,000
CGST	Rs. 2,000
SGST	Rs. 2,000
Gross Invoice Value	Rs. 29,000
Derived TDS Base	Rs. 25,000

The system does not search for a literal field named "TDS Base" because most vendor invoices do not contain such a label. It derives the base from the invoice structure and tax components.

7. Duplicate Invoice Handling

For audit transparency, a duplicate document should remain visible in the TDS Register. However, it should not increase the vendor cumulative amount or TDS liability. The project therefore distinguishes between document visibility and financial effect.

- Original invoice: normal financial effect.
- Duplicate invoice: remains visible and is marked DUPLICATE.
- Duplicate base amount may remain displayed for identification, but its cumulative impact and TDS Now are zero.
- Duplicate detection can use document fingerprint/hash together with vendor, invoice number, date and amount checks.

8. Cumulative TDS Logic

The application is intended to review the current invoice in the context of earlier invoices of the same vendor. The cumulative engine therefore looks beyond the current upload batch and considers invoices already stored in the register for the relevant financial-year/vendor grouping.

- Previous cumulative = eligible base of earlier non-duplicate invoices of the same vendor.
- New cumulative = previous cumulative + current invoice base (except duplicates).
- The TDS Rules master determines the category/section/rate/threshold logic to be applied.
- If a threshold is crossed, the engine can calculate current-period TDS and, where relevant, catch-up TDS on earlier amounts according to the configured rule.
- Vendor matching uses stable identifiers where available and can also rely on confirmed vendor identity to avoid fragmented cumulative chains.

9. Technology and Development Approach

Component	Use in Project
Lovable	Rapid development and deployment of the web application UI and workflow
React / TypeScript	Application interface and business logic
PDF text extraction	Fast reading of text-based invoices
Rule-based parser	Field extraction, numeric validation and amount derivation
Application database	Stores invoices, vendor data, statuses and cumulative values
AI fallback	Used only when deterministic extraction is insufficient, thereby controlling recurring AI cost
Excel export	Provides an office-friendly output for review and further working

10. Key Benefits

- Faster preparation of TDS workings from vendor invoices.
- Reduced manual data entry and reduced risk of missing cumulative vendor history.
- Lower AI operating cost because normal invoices can be processed without an AI call.
- Audit trail through visible invoices, duplicate flags and review statuses.
- Configurable tax rules allow the compliance logic to be maintained separately from document parsing.
- Simple interface suitable for a CA office user who may not be technically trained.
- Scalable from single-invoice upload to multi-invoice / folder processing.

11. Testing Performed

Test Case	Expected Behaviour
Professional fee invoice with one service amount	Correct invoice no., vendor and base amount are extracted
Description containing episode/reference numbers	Reference numbers are not mistaken for money
Invoice containing bank account number	Bank account is ignored for TDS-base selection
Hire invoice with CGST and SGST	Base is separated from GST and gross
Same vendor across different invoice months	Earlier stored invoices contribute to cumulative calculation
Duplicate invoice uploaded again	Second record remains visible as DUPLICATE with zero financial effect
Batch upload	Invoices are processed independently and register is updated without requiring repeated manual entry

12. Limitations and Controls

- Invoice formats are not standard. Some scanned or image-only PDFs may require OCR or AI fallback.
- Tax section, rate and threshold rules can change; therefore the TDS Rules master requires professional maintenance and periodic review.
- Automatic classification is an assistance tool, not a substitute for professional judgement in ambiguous transactions.
- Vendor master data should be periodically reviewed to ensure PAN/GSTIN and default payment classifications are correct.
- Before production use, a broader regression test set covering varied invoice formats should be maintained.

13. Future Scope

- OCR for scanned invoices and photographs.
- Direct import from email attachments or accounting software.
- Vendor master with PAN/GSTIN validation and user-confirmed default TDS treatment.
- Automated TDS payable summary and challan/return working support.
- Exception dashboard for missing PAN, lower-deduction certificates, non-resident payments and special-rate cases.
- Role-based access, audit logs and approval workflow for multi-user CA firms.

14. Conclusion

The AI TDS Compliance Assistant demonstrates how AI and deterministic automation can be combined in a practical Chartered Accountancy workflow. Instead of making AI responsible for every invoice, the project uses fast rule-based extraction for standard documents and reserves AI for exceptions. This makes the solution more economical, explainable and suitable for repeated office use.

The project converts invoice-level data into a structured TDS register, retains duplicate visibility, tracks cumulative vendor exposure and supports professional review through configurable TDS rules. It therefore represents a practical use of AI-assisted automation for tax compliance rather than a purely theoretical AI demonstration.